

Computer Network V.IMP Questions

(EduShine Classes – Arman Ali)

Unit – 1 Introduction & Physical Layer

1. Describe different categories of network. (V.V.IMP)
2. Describe OSI reference model in detail.(V.V.VIMP)
3. Explain functionalities of every layer in OSI reference model with neat block diagram.
4. List the layers in the TCP/IP model.(V.VIMP)
5. What do you mean by topology? Explain in brief any three such network topologies.(V.V.IMP)
6. Write a short note on digital encoding.
7. Encode the data-stream 10011010 using the following encoding scheme:
 - I. Unipolar
 - II. Bipolar NRZ-L
 - III. Bipolar NRZ-I
 - IV. RZ
 - V. Manchester
 - VI. Differential Manchester
 - VII. AMI
8. What do you understand by transmission impairment?

Unit – 2 Link Layer

1. Describe hamming code. How it is used for error detection and correction? Illustrate with the help of suitable example.(V.VIMP)

2. Explain error control mechanism in Data link layer and give example of each method.(IMP)
3. Sender's data $D = 11010$, CRC generator polynomial $= x^4 + x + 1$. Apply CRC algorithm and perform calculations both at sender and receiver end.(V.V.V.V.IMP)
4. A bit stream 10011101 is transmitted using $x^3 + 1$ generator polynomial. Generate the CRC code word for this message.(V.V.V.IMP)
5. State drawbacks of stop and wait protocols.
6. Illustrate the performance issues for Go-Back-N data link protocol.
7. Calculate the total number of transmissions that are required to send 10 data packets through GBN-S and every 5th packet is lost.
8. Explain noisy channel protocols with explanation and diagram.(V.IMP)
9. Explain all random-access protocols.
10. Define any 5 LAN standards.(V.IMP)

Unit – 3 Network Layer

1. Write a short note on internetworking.
2. What is IP addressing? How is it classified? How is subnet addressing performed?(V.V.VIMP)
3. How is the BOOTP different from DHCP?
 - What is the purpose of the domain name system?
 - Discuss the three main divisions of the domain name space.
4. What do you mean by Internet Control Message Protocol (ICMP)?
5. Discuss each command in detail used in networking: (IMP)

- ipconfig
 - netstat
 - ping
 - hostname
 - traceroute
6. Explain all interdomain and intradomain routing algorithms.(V.VIMP)
 7. What is unicast routing? Discuss unicast routing protocols.
 8. What do you mean by congestion control and QoS? What are the parameters of QoS? Explain.(V.VIMP)
 9. The IP network 200.198.160.0 is using subnet mask 255.255.255.224. Design the subnets.(V.V.V.V.IMP)
 10. Illustrate the difference between IPv4 and IPv6.
 11. Define IPv4 header format in detail.(V.IMP)

Unit – 4 Transport Layer

1. Enumerate how the transport layer ensures that a complete message arrives at the destination and in the correct order.
2. Illustrate the header format of UDP.
3. Explain about the TCP header and working of TCP protocol and differentiate between TCP and UDP with frame format.(V.VIMP)
4. Differentiate TCP and UDP in context of the header format.(V.IMP)
5. Define TCP features and header in detail.
6. The following is the dump of a TCP header in hexadecimal format: (V.V.V.IMP)
65320017 00000001 000000 500207FF 00000000
 - a. What is the sequence number?

- b. What is the destination port number?
 - c. What is the acknowledgment number?
 - d. What is the window size?
7. Explain the three-way handshaking protocol to establish the transport level connection.(V.V.IMP)
 8. Explain TCP congestion control algorithm in internet. What is TCP segment header? Also, discuss TCP connection release.
 9. What do you understand by Quality of Service (QoS)? Mention the techniques to improve QoS.
 10. How is TFTP different from FTP?
 11. What three functions can SNMP perform to manage network devices?

Unit – 5 Application Layer

1. Write a short note on DNS in the internet.
2. Compare and contrast SMTP and HTTP.
3. Explain about email architecture and services.
4. Write a short note on file transfer protocol.
5. Write a short note on file transfer protocol. Explain the following application layer protocols : (V.V.IMP)
 - FTP
 - SMTP
 - SNMP
 - DNS
6. Explain the following terms: (V.V.IMP)
 - FTP
 - SMTP
 - DNS
 - ARP
7. Define SNMP protocols and working scenario.
8. Differentiate between FTP and HTTP.(V.IMP)

9. Distinguish between symmetric and asymmetric key cryptography.
10. Explain cryptography. Write about RSA algorithm with an example.(V.VIMP)

Solution Link - [Click here](#)





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BTECH
(SEM VI) THEORY EXAMINATION 2021-22
COMPUTER NETWORKS

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If you require any missing data, then choose suitably.

SECTION A

1. Attempt all questions in brief.**2*10 = 20**

Qno	Questions	CO
(a)	Discuss about transmission mediums in networking.	1
(b)	What do you understand by network topologies.	1
(c)	Explain transmission delay in flow control.	2
(d)	Write a note on round trip time (RTT) in networking.	2
(e)	Discuss the role logical addressing.	3
(f)	Define datagrams in switching.	3
(g)	Discuss about the IP ranges of Class A, B, C and D.	4
(h)	List out prime three functionality of transport layer.	4
(i)	Explain the use of RST flag in TCP header.	5
(j)	Explain HTTP.	5

SECTION B

2. Attempt any three of the following:**10*3 = 30**

Qno	Questions	CO
(a)	Discuss encoding types in physical layer of ISO-OSI model.	2
(b)	Discuss each command in detail used in networking: a- ipconfig b- netstat c- ping d- hostname e- tracert	2
(c)	List out and discuss the disadvantages in STOP N WAIT protocol.	3
(d)	Calculate the total number of transmissions that are required to send 10 data packets through GBN-3 and every 5th packet is lost.	3
(e)	Discuss in detail about ICMP role in network layer.	4

SECTION C

3. Attempt any one part of the following:**10*1 = 10**

Qno	Questions	CO
(a)	Define the relationship between transmission delay and propagation delay, if the efficiency is at least 50% in STOP N WAIT protocol.	2
(b)	Find out window size and minimum sequence number in sliding window protocol, if Transmission delay (T _t)= 1 ms, Propagation delay (T _p)= 24.5 ms. (ms= milliseconds).	2



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BTECH
(SEM VI) THEORY EXAMINATION 2021-22
COMPUTER NETWORKS

4. Attempt any *one* part of the following: 10*1 = 10

Qno	Questions	CO
(a)	Explain distance vector routing (DVR) with working example in detail.	3
(b)	Sender's data D=11010, CRC generator polynomial= x^3+x+1 . Apply CRC algorithm and perform calculations both at sender and receiver end.	3

5. Attempt any *one* part of the following: 10*1 = 10

Qno	Questions	CO
(a)	Assume we want to send a data from S to R and there are 2 routers in between. What will be the total time taken if total number of packets are 5. Data is like: Tp=0 ms, Data size=1000 bytes, BW=1 mbps, Header of the packet=100 bytes.	2
(b)	Explain CSMA/CD in detail.	2

6. Attempt any *one* part of the following: 10*1 = 10

Qno	Questions	CO
(a)	Divide the network with IP address 200.1.2.0 into 5 subnets.	4
(b)	Describe the role of application layer and session layer of OSI model in detail.	4

7. Attempt any *one* part of the following: 10*1 = 10

Qno	Questions	CO
(a)	Write detailed note on "TCP vs UDP".	5
(b)	Explain following application layer protocols: • FTP • SMTP • DNS	5

B.TECH.
(SEM VI) THEORY EXAMINATION 2022-23
COMPUTER NETWORKS

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief. 2 x 10 = 20

- a. Explain the differences between point to point and point to multipoint.
- b. Define bit rate and baud rate.
- c. Compare OSI with TCP/IP protocol suit.
- d. Construct the Polar NRZ-L and NRZ-I schemes for the following Data: 01001110
- e. Describe piggybacking?
- f. Explain ICMP BGP protocol and its application in real-world scenarios.
- g. If a 7-bit hamming code received as 1110101, show that the code word has error. Also, rectify error in this code.
- h. Define QoS.
- i. State difference between HTTP and HTTPS.
- j. Describe the “count to infinity problem” with an example.

SECTION B

2. Attempt any three of the following: 10x3=30

- a. Name and Explain 4 network devices, and write about transmission Impairment in brief.
- b. Explain CSMA/CD with CSMA/CA with diagram.
- c. Explain the working principle of the Congestion Control mechanism with a well-labeled diagram.
- d. Explain the following terms by taking real-world examples:
 - i) Go Back-N
 - ii) Selective repeat.
- e. Explain Asymmetric cryptography. Also, write the steps used in RSA algorithm, demonstrate the transmission of character “F” using RSA.

SECTION C

3. Attempt any one part of the following: 10x1=10

- a. Describe all the layers of the OSI model with a well-labeled diagram.
- b. Differentiate between various topologies with well labeled diagram.

4. Attempt any *one* part of the following: 10x1=10

- a. A bit stream 10011101 is transmitted using x^3+1 generator polynomial. Generate the CRC code word for this message.
- b. Explain error control mechanism in Data link layer and giving example of each method.

5. Attempt any *one* part of the following: 10x1=10

- a. Illustrate the difference between IPv4 and IPv6.
- b. The IP network 200.198.160.0 is using subnet mask 255.255.255.224. Draw the subnets.

6. Attempt any *one* part of the following: 10x1=10

- a. Explain the following terms:
(i) FTP (ii) SMTP (iii) DNS (iv) ARP
- b. Differentiate TCP and UDP in context of the header format.

7. Attempt any *one* part of the following: 10x1=10

- a. Explain DNS.
- b. Define SNMP Protocols and working scenario.



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BTECH
(SEM VI) THEORY EXAMINATION 2023-24
COMPUTER NETWORKS

TIME: 3 HRS

M.MARKS: 100

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2 x 10 = 20

a.	What are the applications of Computer Networks?	02
b.	Write advantages and disadvantages of Mesh Topology.	02
c.	Define fixed and variable size framing.	02
d.	Explain the working of Bridge.	02
e.	Define delivery in Network Layer.	02
f.	Define Token Bucket.	02
g.	Write about Registered ports.	02
h.	Differentiate between connection less and connection oriented.	02
i.	What is Telnet?	02
j.	Define Compression.	02

SECTION B

2. Attempt any three of the following:

3 x 10 = 30

a.	Define TCP/IP Model in detail.	10
b.	Explain Noisy channel Protocols with explanation and diagram.	10
c.	Define IPv4 header Format in detail.	10
d.	Define TCP features and header in detail.	10
e.	Explain SNMP in detail.	10

SECTION C

3. Attempt any one part of the following:

1 x 10 = 10

a.	Name and explain Network Devices on each layer of OSI Model.	10
b.	Define various topologies with diagram, features, and pros-cons.	10

4. Attempt any one part of the following:

1 x 10 = 10

a.	Define any 5 LAN standards.	10
b.	Explain all Random-Access protocols.	10

5. Attempt any one part of the following:

1 x 10 = 10

a.	Explain all Interdomain and Intradomain routings algorithms.	10
b.	Senders' data D= 110101, CRC generator polynomial= x^3+x+1 . Apply CRC algorithm and Perform calculations both at sender and receiver end.	10

6. Attempt any one part of the following:

1 x 10 = 10

a.	Define Connection Management. Explain how Three-way handshaking using diagram.	10
b.	Write about Quality-of-Service Parameters.	10

7. Attempt any one part of the following:

1 x 10 = 10

a.	Differentiate between FTP and HTTP.	10
b.	Explain Cryptography. Write about RSA Algorithm with example.	10